Qwen

# From Raw Errors to Hard Governance: An Empirically Calibrated, Mathematically Provable Framework for Data Quality Risk

## Empirical Calibration of Ingest Quality Risk via `rcalib`

The calibration of the ingest quality risk metric, `rcalib`, represents the foundational layer upon which the entire data quality governance framework is built. Its purpose is to translate raw, row-level diagnostic events within an ingest window into a single, normalized risk coordinate, `rcalib`, that accurately reflects the severity of data integrity issues . This process prioritizes empirical grounding, leveraging historical Phoenix `qpudatashard` data to determine the critical parameters of the system: the weights assigned to different types of ingest errors and the boundaries of the corridor bands that map the aggregate error count into a risk score . This empirical-first approach acknowledges that the largest source of residual risk lies not in abstract theory but in the precise geometry of how real-world failures are mapped onto the risk scale . The strategy involves using regression analysis on historical ingest logs to fit the parameters so that `rcalib` correlates strongly with observed downstream degradation in performance metrics like `K` (Knowledge) and `E` (Efficacy), as well as documented incident logs .

The core of the `rcalib` calculation begins with the construction of an ingest error scalar, `I`. This scalar is a weighted sum of various diagnostic counts accumulated over a closed ingest window $[t_0, t_1]$ . There are two forms of this kernel. The minimal form, used for initial assessments, relies on diagnostics directly tied to the `EcoNetSchemaShard2026v1` specification. It defines four key counts: $N_{miss}$ for missing mandatory fields, $N_{rfc}$ for rows violating strict RFC-4180 CSV grammar, $N_{type}$ for values that cannot be parsed into their declared types, and $N_{varid}$ for headers referencing unknown variable IDs not present in the active `CorridorTable` . These counts are combined into the scalar `I_rcalib` using a set of hazard-weighted coefficients:

$$I_{\text{rcalib}} = w_{\text{miss}} N_{\text{miss}} + w_{\text{rfc}} N_{\text{rfc}} + w_{\text{type}} N_{\text{type}} + w_{\text{varid}} N_{\text{varid}}$$

In this formulation, the weights $w_{miss}$ and $w_{rfc}$ are assigned to the highest hazard band, reflecting their severe impact on data integrity, while $w_{type}$ is assigned a medium weight and $w_{varid}$ a low weight, mirroring a predefined CEIM hazard ordering . For more comprehensive analysis, the full CEIM ingest kernel expands this to a more general scalar `I` using weights `m, s, c, u` and a broader set of diagnostic counts: $N_{missing}$ (missing required columns/fields), $N_{schema}$ (rows violating the structural spec of `EcoNetSchemaShard2026v1`), $N_{corridor}$ (references to varids with no corresponding `CorridorBands` row), and $N_{unit}$ (inconsistencies in measurement units) [7] . The choice of weights (`m,s,c,u`) is therefore a critical calibration parameter, and the recommended approach is to use Phoenix ingest logs to regress these values against actual downstream consequences, ensuring that a unit of one error type (e.g., schema violation) reflects a larger marginal risk than another (e.g., unit inconsistency), grounded in empirical failure patterns .

Once the ingest error scalar `I` is computed, it must be mapped into the normalized risk coordinate `rcalib` ∈ [0,1]. This is achieved through a standard corridor-normalization kernel, which uses a set of empirically determined corridor bands stored in the `ecosafety.corridors.v2` table for the `varid = rcalib` row [31] . These bands are defined as `(Isafe, Igold, Ihard)` and correspond to the `safemin/safemax`, `goldmin/goldmax`, and `hardmin/hardmax` values in the table, along with an associated risk level at the gold band, `rgold` ∈ (0,1) . The explicit piecewise mapping is as follows: - If $I \le I_{safe}$, then $r_{calib}=0$. This represents a deeply safe ingest operation. - If I_{safe} < I \le I_{gold} , then $r_{calib}$ ramps linearly from 0 to $r_{gold}$ according to the formula:

$$r_{\text{calib}} = \frac{I - I_{\text{safe}}}{I_{\text{gold}} - I_{\text{safe}}} \cdot r_{\text{gold}}$$

- If I_{gold} < I < I_{hard} , then $r_{calib}$ continues to ramp linearly from $r_{gold}$ up to 1:

$$r_{\text{calib}} = r_{\text{gold}} + \frac{I - I_{\text{gold}}}{I_{\text{hard}} - I_{\text{gold}}} \cdot (1 - r_{\text{gold}})$$

- If $I \ge I_{hard}$, then $r_{calib}=1$. This signifies chronic ingest failure .

The calibration of the corridor boundaries $(I_{safe}, I_{gold}, I_{hard})$ is a crucial empirical task. The target distributions for `rcalib` serve as guiding principles: typical clean operation should yield a `rcalib` value in the range of approximately 0–0.1, moderate ingest problems should fall into the 0.1–0.3 range, and chronic failures should be pushed near

1 . To achieve this, analysts can compute the empirical distributions of the ingest error scalar `I` for windows that have been manually labeled as "clean," "moderately problematic," and "chronic fault" (using incident logs or high `rcalib` cases as a proxy). By choosing the band positions, they can ensure that the quantiles of these clusters map into the desired target ranges for `rcalib` . For instance, `Isafe` would be chosen as the 99th percentile of the "clean" distribution, `Igold` as the 75th percentile of the "problematic" distribution, and `Ihard` as the 95th percentile of the "chronic" distribution, adjusted through iterative regression to match the target `rcalib` bands . In practice, a simplified three-segment kernel that maps `I <= Isafe` to 0 and then applies a single linear ramp to 1 by `Ihard` is often used, with the slope and implicit gold position chosen by regression over ingest-failure logs to meet the same target distributions .

This entire process—from the weighted sum of counts to the piecewise affine normalization—is designed to possess certain provable mathematical properties that are essential for robust governance. First, due to the use of non-negative weights, the ingest error scalar `I` is a monotone non-decreasing function of each individual diagnostic count . Since the corridor-normalization kernel is also piecewise linear and non-decreasing in `I`, it follows that `rcalib` is a monotone non-decreasing function of each underlying diagnostic count. This property formalizes the intuitive requirement that increasing the number of any type of ingest fault can never decrease the calculated risk, thus preventing the masking of degrading data quality. Second, because `I` is linear in the counts and the normalization kernel is piecewise linear, `rcalib` is a convex function of the diagnostic counts on each segment of the piecewise map . This convexity ensures that averaging ingest histories across multiple windows is a conservative estimate of risk; it prevents the situation where averaging a "good" window with a "bad" window could produce an artificially low risk score, thereby hiding significant spikes in ingest error. This property supports the argument that chronic small errors should be treated as seriously as occasional large spikes . These empirical calibration efforts, when paired with these mathematical guarantees, create a system where `rcalib` is not just a diagnostic tool but a reliable and verifiable measure of ingest quality risk. The final step in operationalizing this is to implement the logic in a production-ready module, such as the provided C++ `ingest_rcalib.hpp` file, which deterministically computes `I` and `rcalib` from the diagnostic counts and corridor parameters . Treating this crate as the single source of truth for all ingest-quality computations removes the risk of divergent re-implementations and solidifies the governance foundation .

| Component | Description | Source Data / Parameters | Calibration Method |
|---|---|---|---|
| Diagnostic Counts | Row-level event counters for data integrity issues. | `qpudatashard` diagnostics (e.g., `Nmissing`, `Nschema`, `Ncorridor`, `Nunit`). | Not applicable; these are direct measurements. |
| Ingest Error Weights | Scalar multipliers for diagnostic counts, ordered by hazard. | `m, s, c, u` or `wmiss, wrfc, wtype, wvarid`. | Regression on historical ingest logs to correlate with downstream `K/E/R` degradation. |
| Corridor Bands | Thresholds `(Isafe, Igold, Ihard)` for mapping scalar `I` to risk. | `safemin/max`, `goldmin/max`, `hardmin/max` in `ecosafety.corridors.v2`. | Fitting band positions to place empirical `I` distributions of labeled "clean", "moderate", and "chronic" windows into target `rcalib` bands (0-0.1, 0.1-0.3, >0.3). |
| Normalization Kernel | Piecewise affine function mapping `I` to `rcalib` ∈ [0,1]. | Defined by corridor bands and `rgold`. | Regression to choose slopes and `rgold` so that typical clean operation sits near `rcalib` ≈ 0–0.1, moderate problems in 0.1–0.3, and chronic failures near 1. |
| Final Output (`rcalib`) | Normalized ingest quality risk score. | Computed from `I` and the normalization kernel. | Verified via CI replay jobs that test monotonicity under synthetic ingest faults. |

## Sensor Uncertainty Quantification through the `rsigma` Composite

Parallel to the calibration of ingest quality, the `rsigma` metric provides a rigorous framework for quantifying sensor uncertainty, elevating it from a secondary concern to a first-class citizen in the system's safety and governance architecture. The objective is to construct a composite risk score, `rsigma`, that aggregates various modes of sensor degradation—such as drift, noise, and bias—into a single, interpretable value in the `[0,1]` interval . This score is then integrated into the Lyapunov residual, ensuring that uncertainty in the underlying physical measurements is explicitly accounted for in the overall system risk assessment. The recommended approach is to build `rsigma` using a weighted L2 norm of its constituent components, a choice driven by its desirable mathematical properties, including smoothness and convexity, which integrate cleanly with the existing quadratic Lyapunov framework . Alternative norms, such as L1 or L∞, may be explored as secondary diagnostic tools if specific failure patterns suggest they offer better predictive power .

The construction of `rsigma` begins with identifying and normalizing its individual component uncertainties. These components are defined per sensor family (e.g., flow, pressure, turbidity, temperature) and represent distinct modes of data degradation . The standard components are `rdrift`, representing the magnitude of long-term sensor drift;

`rnoise`, representing short-term variability or noise (often measured as standard deviation over a sliding window); `rbias`, representing systematic deviation from a reference standard; and `rloss`, an optional coordinate for data loss or periods of low sample density . For each of these raw metrics, a separate set of corridor bands, derived from historical calibration logs and manufacturer performance specifications, is used to normalize them into a `[0,1]` risk coordinate . For example, a raw drift measurement would be mapped into `r_drift` using a piecewise affine kernel similar to the one used for `rcalib`, based on safe, gold, and hard thresholds for drift specific to that sensor type. This process ensures that each uncertainty mode is scaled consistently and meaningfully before being combined into the composite score.

The primary formula for the composite `rsigma` is a weighted Euclidean norm, clamped at 1 to maintain it within the `[0,1]` interval:

$$r_\sigma = \min\left(1, \sqrt{\alpha_{\text{drift}} r_{\text{drift}}^2 + \alpha_{\text{noise}} r_{\text{noise}}^2 + \alpha_{\text{bias}} r_{\text{bias}}^2 + \alpha_{\text{loss}} r_{\text{loss}}^2}\right)$$

The weights $\alpha_k$ reflect the relative hazard of each uncertainty mode, with drift and bias typically assigned higher weights than noise and loss, as they often have a more profound impact on control decisions and ecological interpretations . The selection of these weights is a critical calibration step. The recommended methodology involves analyzing joint distributions of `rsigma` and observed downstream physics failures or bad eco-impact events . By regressing these failure indicators against the individual component risks ($r_{drift}$,$r_{noise}$,$etc.$), one can derive weights that best explain the variance in negative outcomes, aligning the composite score with real-world sensor reliability concerns . The resulting `rsigma` value is itself placed within a set of corridor bands—typically defined as "deeply safe" (0–0.10), "gold" (0.10–0.30), and "hard" (>0.30)—which are also calibrated from sensor performance data to define acceptable versus unacceptable levels of uncertainty .

The mathematical properties of this L2-norm formulation are central to its utility. First, since the weights $\alpha_k$ are non-negative and each component risk $r_k$ is bounded in `[0,1]`, the quantity inside the square root is non-negative, ensuring the expression is always real-valued. The subsequent min(1, ·) operation guarantees that the final `rsigma` score remains firmly within the `[0,1]` range, making it directly comparable to other risk coordinates like `rcalib` and the physical risk variables . Furthermore, `rsigma` is a monotone non-decreasing function of each of its component risks. An increase in drift, noise, or bias will either increase `rsigma` or leave it unchanged, but never decrease it. This property is fundamental to the "no inflation" principle, ensuring that increased

sensor uncertainty is never hidden. Most importantly, the L2-norm structure makes `rsigma` a 1-Lipschitz continuous function of its inputs . This means that the rate of change of `rsigma` with respect to any single component risk is bounded. Small changes in a sensor's raw output (e.g., a slight increase in noise) cannot cause arbitrarily large jumps in the `rsigma` score. This Lipschitz continuity is crucial for numerical stability and for bounding the change in the Lyapunov residual $V_t^{full}$, which is necessary for proving the system's stability properties .

Just as with `rcalib`, the calibration of `rsigma`'s component bands and weights is an empirical endeavor rooted in historical Phoenix data. The process involves gathering extensive sensor calibration histories and residuals (measures of drift, noise, bias, and loss) for each sensor family deployed in the Phoenix network . Analysts can then correlate these raw uncertainty metrics with known "bad sensor" events recorded in maintenance logs or with periods of anomalous physical readings. By fitting the component corridor bands and weights, they can ensure that the `rsigma` score reliably flags impending failures and distinguishes between benign fluctuations and serious degradation . For instance, a small, temporary spike in `rnoise` might push `rsigma` into the "gold" band, while sustained `rbias` or `rdrift` would drive it toward the "hard" band. The Lyapunov weight $w_\sigma$, drawn from the `rsigma` row in `ecosafety.corridors.v2`, determines the sensitivity of the overall residual `Vt_full` to changes in sensor uncertainty . Co-tuning $w_\sigma$ with the component weights ensures that data quality risk carries appropriate weight relative to physical risk in the governing equations . Implementing this in a production environment, for example in the Rust `rsigma` workload module, requires reading sensor statistics from `qpudatashards`, computing the component risks, calculating the composite `rsigma`, and writing the result to ecosafety shards alongside `rcalib` . This creates a unified stream of data quality information that can be audited and governed as a cohesive entity.

| Component | Description | Source Data / Parameters | Calibration Method |
|---|---|---|---|
| Component Uncertainties | Per-sensor-family raw metrics of degradation. | Sliding window stats (drift, noise), lab comparisons (bias), data gaps (loss) from `qpudatashards`. | Using sensor calibration logs and residuals to define safe/gold/hard bands for each component. |
| Normalized Component Risks | `[0,1]` scores for each uncertainty mode. | `rdrift`, `rnoise`, `rbias`, `rloss` computed via piecewise affine corridor kernels. | Calibrated by mapping raw component values to `[0,1]` using component-specific corridor bands. |
| Composite Weight $\alpha_k$ | Relative importance of each uncertainty mode. | $\alpha_{drift}, \alpha_{noise}, \alpha_{bias}, \alpha_{loss}$. | Regression on downstream incidents and K/E/R degradations to determine weights that best predict failures. |
| `rsigma` Corridor Bands | Thresholds for the composite uncertainty score. | Safe (e.g., 0–0.10), Gold (0.10–0.30), Hard (>0.30) bands in `ecosafety.corridors.v2`. | Aligning `rsigma` distributions with real-world failure modes and false positive rates. |
| Final Output (`rsigma`) | Normalized sensor uncertainty risk score. | Computed from component risks and weights via the L2-norm formula. | Verified via CI replay jobs that inject synthetic uncertainty (e.g., artificial drift/noise) and check for expected increases in `rsigma` and `Vt_full`. |

# Mathematical Foundations: Monotonicity and Lyapunov Stability Guarantees

The integration of `rcalib` and `rsigma` into the governance framework is not merely a matter of adding new metrics; it requires establishing a firm mathematical foundation to guarantee the system's behavior is predictable, stable, and secure. The core of this foundation rests on two pillars: proving the monotonicity of the risk metrics with respect to their inputs and verifying that the system adheres to a discrete-time Lyapunov stability condition. These proofs are not abstract exercises; they are intended to be developed in lockstep with the empirical calibration of corridor bands and weights, providing immediate, formal validation of the system's design . They are encoded as short lemmas in documentation and, crucially, as executable ALN invariants that can be checked by the system's Continuous Integration (CI) pipeline, effectively turning the proofs into automated tests that prevent regressions . This dual approach of empirical fitting and formal verification ensures that the framework is both practically effective and theoretically sound.

The first pillar, monotonicity, ensures that the system's perceived risk never decreases when the underlying conditions worsen. For `rcalib`, this property arises naturally from its construction. Given that the ingest error scalar `I` is a weighted sum of diagnostic counts with non-negative weights, and the normalization kernel is piecewise linear with

non-negative slopes, `rcalib` is a monotone non-decreasing function of each individual diagnostic count $(N_{missing}, N_{schema}, etc.)$ . Formally, if a window's diagnostic counts increase or remain the same, its `rcalib` score cannot decrease. Similarly, for `rsigma`, which is defined as a weighted L2 norm of its component risks, and given that the weights $\alpha_k$ are non-negative, `rsigma` is a monotone non-decreasing function of each component risk $(r_{drift}, r_{noise}, etc.)$ . An increase in any component, caused by a deterioration in the corresponding raw sensor metric (e.g., more drift), will either increase `rsigma` or leave it unchanged. This property is fundamental to the "only down-credit, never inflate" principle. It guarantees that as data quality degrades—whether through more ingest errors or greater sensor uncertainty—the system's risk score will not falsely improve, thus preventing the masking of deteriorating conditions. These monotonicity properties can be formally stated in ALN invariants, for example: "If any of the ingest error counts increase, then the resulting `rcalib` must be greater than or equal to the original `rcalib`" .

The second and more profound pillar is the Lyapunov stability condition, which governs the evolution of the system's total risk over time. The foundation for this is the extended Lyapunov residual, $V_t^{full}$ . This scalar quantity aggregates all sources of risk—including physical processes, ingest quality, and sensor uncertainty—into a single energy-like function:

$$V_t^{\text{full}} = \sum_{j \in \text{physical}} w_j r_{j,t}^2 + w_{\text{calib}} r_{\text{calib},t}^2 + w_\sigma r_{\sigma,t}^2$$

This formulation elegantly combines the base residual on physical risk coordinates $r_{j,t}$ with the newly introduced data quality terms, `rcalib` and `rsigma`, each multiplied by its own Lyapunov weight $(w_{calib}, w_\sigma)$ taken directly from the `ecosafety.corridors.v2` table . The non-negativity of all weights and the fact that every risk coordinate $r \in [0,1]$ ensures that $V_t^{full}$ is a non-negative, convex quadratic form. It equals zero only when all risk coordinates simultaneously vanish, making it a valid candidate for a Lyapunov function . The core safety guarantee is the `safestep` invariant, which is encoded as an ALN lemma stating: "For all accepted steps, either the residual at the next timestep is less than or equal to the current residual $(V_{t+1}^{full} \leq V_t^{full})$ or the system is already operating within a deeply safe interior region $(V_t^{full} \leq V_{int})$" . Here, $V_{int}$ is a small pre-defined threshold that demarcates this safe interior . This discrete-time Lyapunov condition ensures that the system's risk trajectory is non-expansive outside the safe interior, providing a formal mechanism for a controller to enforce stability and prevent runaway risk accumulation [14] [113].

The inclusion of `rcalib` and `rsigma` preserves the desirable stability properties of this invariant. Because `rcalib` and `rsigma` are themselves monotone non-decreasing functions of their inputs, and because their contributions to $V_t^{full}$ are quadratic and weighted by non-negative coefficients, any increase in ingest error or sensor uncertainty will weakly increase the Lyapunov residual . Specifically, the partial derivative of $V_t^{full}$ with respect to `I` can be shown to be non-negative wherever the normalization kernel is ramping, supporting the intuition that "more ingest error cannot lower the Lyapunov residual" . This property is what allows the system to treat poor data quality as a genuine threat to system stability. A controller enforcing the `safestep` rule will take corrective action whenever `V_t` threatens to exceed $V_{int}$, and a rising `rcalib` or `rsigma` is one of the primary triggers for such an action. The proofs demonstrate that tightening the corridor bands (i.e., making them narrower) for `rcalib` or `rsigma` is equivalent to pointwise increasing the risk score for a fixed input error, which in turn monotonically increases the Lyapunov residual. This proves that policy tightening in response to improved practices is a conservative and verifiable improvement, as it only serves to make the system's safety gates harder to pass, not easier . These mathematical foundations, when codified as ALN invariants and validated against CI replay jobs, transform the governance framework from a heuristic model into a provably correct system where safety is not just a goal, but a guaranteed property of its execution.

| Proof Obligation | Statement | Mathematical Justification | Implementation |
|---|---|---|---|
| **Monotonicity of `rcalib`** | Increasing any diagnostic count ($N_{missing}$,$N_{schema}$,...) cannot decrease `rcalib`. | `rcalib` is a composition of a linear, non-decreasing function `I` and a piecewise affine, non-decreasing normalization kernel. | ALN Invariant: "If `N_missing' >= N_missing`, then `rcalib' >= rcalib`." CI Replay Job: Inject synthetic counts and verify monotonicity. |
| **Convexity of `rcalib`** | Averaging good and bad ingest windows does not produce an artificially low average `rcalib`. | `rcalib` is a convex function of the diagnostic counts on each linear segment of its piecewise definition. | Used to justify window-averaging for risk estimation. |
| **Boundedness of `rsigma`** | `rsigma` is always in the `[0,1]` interval. | The weighted L2 norm is non-negative, and the `min(1, ·)` operation clamps the result at 1. | Ensures compatibility with other risk coordinates and the `R` metric. |
| **Lipschitz Continuity of `rsigma`** | Small changes in sensor data do not cause arbitrarily large changes in `rsigma`. | The L2 norm is a 1-Lipschitz function, ensuring `rsigma` is well-behaved for stability analysis. | Essential for bounding the change in the Lyapunov residual `ΔVt`. |
| **Lyapunov Property of `Vt_full`** | `Vt_full` is a valid discrete Lyapunov candidate function. | `Vt_full` is a sum of non-negative quadratic terms, making it a non-negative, convex function that is zero only at the origin. | Forms the basis for the `safestep` controller logic. |
| **Stability under `safestep`** | The `safestep` invariant prevents risk expansion outside the safe interior. | The controller enforces `Vt+1 <= Vt` if `Vt > Vint`, ensuring the risk trajectory is non-increasing until it enters the safe region. [14] [113] | Encoded as an ALN invariant and enforced by the controller. |
| **No-Masking of High Risk** | High underlying risk (`rcalib` or `rsigma`) is never hidden by trust scalars. | `R` is defined as the max over all coordinates, including un-scaled `rcalib` and `rsigma`. Trust scalars only multiply `K` and `E`. | Proven by the definition of `R` and `E=1-R`. |
| **Non-Inflation of Adjusted Scores** | `K_adj` and `E_adj` are non-increasing functions of `rcalib`. | With `D_data = 1 - rcalib`, the partial derivatives `∂K_adj/∂rcalib` and `∂E_adj/∂rcalib` are non-positive. | Provides the mathematical backbone for the "only down-credit" principle. |

# Governance Architecture: The Dual-Control KER-R Framework

The ultimate purpose of calibrating `rcalib` and `rsigma` and establishing their mathematical foundations is to embed them into a robust governance architecture capable of driving sound operational decisions. The proposed framework extends the existing Knowledge-Efficacy-Risk (KER) paradigm to explicitly account for data quality, introducing a dual-control mechanism that balances positive incentives with hard safety constraints. This architecture is designed around a clear distinction between primary evaluation metrics and absolute blocking authorities. The primary metrics, `K_adj`

(adjusted Knowledge) and `E_adj` (adjusted Efficacy), serve as the principal measures for granting rewards, promotions, and eco-credits. In contrast, the maximum risk score, `R`, retains independent hard blocking authority, ensuring that no window can be deemed deployable if its underlying risk, even from data quality sources, exceeds predefined thresholds . This design enforces the "only down-credit, never inflate" principle, creating a system where high-quality data is not just a preference but a non-negotiable prerequisite for participation in the trusted ecosystem.

The extension of the KER framework is straightforward but powerful. Over a given time window of length $N$, the core metrics are recalculated to include the full spectrum of risk coordinates . The Knowledge factor, `K`, is defined as the fraction of timesteps within the window for which the `safestep` condition was met. The maximum risk score, `R`, is updated to be the maximum value of *all* risk coordinates across all timesteps in the window, including every physical process variable ($r_j$) plus `rcalib` and `rsigma`. Consequently, the Efficacy score, `E`, is simply the complement of `R`: $E=1-R$ . This simple yet profound modification ensures that sustained ingest faults or persistent sensor uncertainty will inevitably propagate through the system, causing `R` to increase and `E` to decrease. Even if all physical process variables remain perfectly nominal, a window plagued by chronic data quality issues will be flagged with a high `R` and a correspondingly low `E`, correctly reflecting its compromised state . The `KerWindow.update` logic in the core software must be updated to track the maximum values of `rcalib` and `rsigma` during each window and incorporate them into the computation of `R` .

To further refine the evaluation process, the framework introduces a combined trust scalar, `D_combined`, which modulates the primary metrics. This scalar is the product of two components: the existing Multonry sensor trust scalar, `D_sensor`, which is derived from sensor drift, variance, and residual patterns, and a new data ingest trust scalar, `D_data` . The data trust scalar is defined as a simple linear function of `rcalib`: $D_{data}=1-r_{calib}$ . This formulation directly ties ingest quality to a trust factor in the `[0,1]` interval, where perfect ingest quality yields `D_data = 1` and catastrophic ingest failure yields `D_data = 0`. The combined trust is then $D_{combined}=D_{sensor}\cdot D_{data}$ . Both the raw Knowledge and Efficacy scores (`K_raw`, `E_raw`) are then down-scaled by this combined trust to produce the adjusted metrics:

$$K_{adj}=K_{raw}\cdot D_{combined}, \quad E_{adj}=E_{raw}\cdot D_{combined}$$

These adjusted scores are designated as the primary metrics for all positive evaluations and resource allocations . Their mathematical relationship to `rcalib` provides the

formal proof for the "only down-credit, never inflate" principle. For fixed K_raw, E_raw, and D_sensor, the partial derivatives of K_adj and E_adj with respect to rcalib are:

$$\frac{\partial K_{adj}}{\partial r_{calib}} = -K_{raw} D_{sensor} \leq 0, \quad \frac{\partial E_{adj}}{\partial r_{calib}} = -E_{raw} D_{sensor} \leq 0$$

Since trust scalars are non-negative, these derivatives are always less than or equal to zero . This proves that K_adj and E_adj are strictly monotone non-increasing functions of rcalib. Any degradation in ingest quality, reflected as an increase in rcalib, will necessarily lead to a decrease or, at best, no change in K_adj and E_adj. It is impossible for a window to become more attractive after its ingest quality degrades. This property is the mathematical bedrock of the principle, ensuring that the system cannot be gamed by inflating scores with untrustworthy data.

Despite the down-scaling of K and E, the maximum risk score R remains untouched by any trust adjustments. This is a critical design feature that prevents the masking of high underlying risk. The governance logic for deployment decisions, encapsulated in the KerDeployableConfig and its corresponding ALN invariant, operates on the raw, un-scaled metrics. A decision is made based on whether K_adj is above a minimum threshold, E_adj is above a minimum threshold, and, most importantly, R is below a maximum threshold . The retention of R's hard blocking authority means that a window with exceptionally high rcalib or rsigma—perhaps due to a novel but undetected ingest pipeline failure or a sudden sensor malfunction—will be blocked from deployment, regardless of how attractive its K_adj and E_adj scores might appear. This dual-control mechanism creates a fail-safe system: K_adj and E_adj provide a nuanced, graded reward system for good performance, while R acts as an unassailable gatekeeper, protecting the integrity of the entire ecosystem from high-risk, low-trust operations. This architecture transforms data quality from a passive diagnostic into an active, enforceable governance lever.

## Operationalization and Cross-Plane Coupling

Translating the calibrated risk metrics and governance framework into a fully operational system requires a series of concrete implementation steps, spanning software development, testing, and user-facing tooling. The goal is to create a seamless, auditable, and provably correct workflow where data quality risk is an integral part of every governance decision. This involves implementing the core computational kernels in

production languages like C++ or Rust, establishing a rigorous Continuous Integration (CI) pipeline with automated replay jobs to validate mathematical properties, and designing dashboards that provide operators with actionable insights into data quality. A crucial aspect of this operationalization is demonstrating the cross-plane coupling effect: showing how improving data quality not only benefits ingest but also positively impacts the governance of the entire system by reducing the overall Lyapunov residual and relaxing deployability constraints.

The first step in operationalization is the implementation of the core logic. The provided learning materials outline a C++ layout that mirrors the required functionality, with dedicated header files for `ingest_rcalib`, `rsigma`, and `lyapunov_ker` calculations . These modules should be compiled into a reusable library, `ecosafety-core`, that reads corridor bands and weights from `qpudatashard` tables or ALN particles at startup . This ensures that the parameters are treated as immutable during a run, preserving the integrity of the calibration. Alongside these implementations, a suite of property tests must be written to verify the fundamental mathematical properties. These tests would confirm that `rcalib` and `rsigma` are monotone non-decreasing in their underlying counts and raw uncertainties, and that they always remain within the `[0,1]` bounds . The `vt_full` function in the Lyapunov kernel must be confirmed to correctly aggregate all risk coordinates, including `rcalib` and `rsigma`, using their respective Lyapunov weights . This deterministic implementation serves as the single source of truth, eliminating ambiguity and ensuring that all nodes in the network perform the exact same risk calculations .

With the core logic in place, a robust CI pipeline is essential to maintain the system's integrity. The most critical component of this pipeline is the replay job, which serves as an automated proof of the system's properties . These jobs would take historical Phoenix `qpudatashard` data and synthetically introduce controlled perturbations. For example, a job could be created to inject a known number of extra schema violations or unit errors into a shard and then verify that the resulting `rcalib`, the extended Lyapunov residual `Vt_full`, and the maximum risk `R` are all weakly increasing. Simultaneously, it would verify that the trust-adjusted scores `K_adj` and `E_adj` are weakly decreasing. Any violation of these monotonicity conditions would cause the CI job to fail, automatically blocking the merge of any code or corridor parameter changes that introduce such regressions . This creates a self-correcting feedback loop where empirical results are continuously validated against mathematical guarantees. The hex-stamping of ALN invariants, such as `safestep.aln`, provides cryptographic evidence that the rules governing the system are transparent, immutable, and subject to this automated verification process .

To realize the full potential of the framework, operators and communities need tools to visualize and act upon the data quality signals. A "Phoenix Ingest Quality Dashboard" is a prime example of a valuable tool that can be built on top of this framework . Such a dashboard would read from `IngestRcalibPhoenix2026v1.aln` shards to display `rcalib`, `K_adj`, and `E_adj` per node and per window. By highlighting ingest pipelines whose chronic errors are silently eroding eco-impact, it would guide engineering teams toward targeted fixes that ultimately increase effective pollutant removal and reduce wasted sampling . Similarly, a "Rsigma and Rcalib Forensics Runner" could be used to replay historical system failures, reconstructing the trajectories of `rcalib` and `rsigma` before and after an incident. This forensics capability would help tighten corridor bands where near-misses occurred, driving `R` downward over time without having to relax the more sensitive physical bands . These tools transform the abstract risk scores into concrete, actionable intelligence for system operators.

Finally, the framework's strength lies in its ability to couple different planes of risk under a single, unified mathematical umbrella. The Lyapunov residual `Vt_full` is the vehicle for this coupling. When a corridor for `rcalib` or `rsigma` is tightened—for example, by setting a lower `I_hard` threshold or a smaller `u_hard` band—this change monotonically increases the risk score for a fixed amount of underlying error . As shown by the derivative $\frac{\partial V_t^{full}}{\partial I} = 2w_{\text{calib}} r_{\text{calib}} \frac{\partial r_{\text{calib}}}{\partial I}$ , a tighter corridor leads to a higher `rcalib` for the same `I`, which in turn causes a higher `Vt_full` . This increase in the Lyapunov residual can trigger stricter governance actions, such as derating a lane or blocking a node from promotion, even if the physical process variables have not changed. This demonstrates that improving data quality is not just about fixing the ingest path; it is a systemic intervention that pushes the entire system deeper into a safer operational regime, reduces the overall risk burden, and makes the governance of the physical plane more robust and reliable. This cross-plane effect is a key justification for treating `rcalib` and `rsigma` as first-class citizens in the governance model.

| Action Item | Description | Purpose | Key Components |
|---|---|---|---|
| **Core Module Implementation** | Write deterministic C++ or Rust libraries for `rcalib`, `rsigma`, and `Vt_full` calculations. | Create a single, unambiguous source of truth for all risk computations. | `ingest_rcalib.hpp`, `rsigma.hpp`, `lyapunov_ker.hpp` ; Property tests for monotonicity and bounds . |
| **CI Replay Jobs** | Develop automated jobs that inject synthetic errors/failures into historical shards and verify system responses. | Provide automated, empirical proof of monotonicity and Lyapunov properties; block regressions. | Synthetic injection of `Nmissing`, `Nschema`; verification that `rcalib`, `Vt_full`, `R` weakly increase and `K_adj`, `E_adj` weakly decrease . |
| **ALN Invariant Encoding** | Formalize mathematical proofs and governance rules as ALN invariants with hex-stamps. | Make governance rules transparent, cryptographically verifiable, and executable by the blockchain. | `safestep.aln` for Lyapunov condition; `invariantkerdeployable` for KER gating rules ; Monotonicity lemmas . |
| **Operational Dashboards** | Build user-facing tools to visualize data quality metrics alongside physical metrics. | Translate abstract risk scores into actionable insights for operators and auditors. | Ingest Quality Dashboard ; Rsigma Forensics Runner . |
| **Cross-Plane Coupling Analysis** | Analyze how changes to data quality corridors affect the overall Lyapunov residual and governance outcomes. | Demonstrate the systemic benefit of improving data quality beyond the ingest plane. | Derivative analysis of `Vt_full` w.r.t. corridor parameters ; Simulation of corridor tightening effects. |
| **Unified Shard Schema** | Ensure that `qpudatashard` schemas carry all relevant metrics together in a single, auditable record. | Enable holistic auditing of a window's performance by linking raw data, risk scores, and governance decisions. | `CyboquaticPhoenixEcosafety2026v2.aln` schema carrying `K, E, R, rcalib, rsigma, Vt, DeployDecision` and `evidencehex` . |

# Synthesis and Recommended Research Path

This research report has detailed a comprehensive framework for calibrating and governing data quality risk through the `rcalib` and `rsigma` metrics. The core of the framework is a multi-layered architecture that integrates empirical calibration, mathematical proof, and layered governance. The strategic priority is to begin with the empirical calibration of corridor bands and weights using historical Phoenix `qpudatashard` data, as this phase holds the most significant impact on the system's real-world effectiveness . This empirical work must proceed in lockstep with the implementation of mathematical proofs and ALN invariants to ensure that every calibrated parameter is accompanied by a formal guarantee of its behavior . The framework establishes `rcalib` and `rsigma` as hard, certifiable brakes on the system, with the `rsigma` composite maintaining its current weighted L2 norm as the primary

governance scalar due to its favorable convex properties . The governance model employs a dual-control mechanism: `K_adj` and `E_adj` serve as the primary metrics for rewards and credits, while the raw maximum risk score `R` retains independent hard blocking authority, enforcing the "only down-credit, never inflate" principle .

The recommended research path forward is structured and actionable. The initial phase must focus entirely on empirical calibration. This involves conducting regression studies on Phoenix ingest logs to fit the weights (`m,s,c,u`) and corridor bands for `rcalib`, aiming to align its distribution with observed downstream K/E/R degradation . Concurrently, a similar study should be performed on sensor calibration histories to calibrate the component bands and weights for `rsigma` . The outputs of these studies should be written to research-lane `qpudatashards` and used to update the corresponding ALN corridor definitions, such as `IngestRcalibPhoenix2026v1.aln` . Immediately following this, the mathematical proofs of monotonicity and Lyapunov stability must be formalized. Short, explicit inequalities and lemmas should be documented in the `ecosafety-core` documentation and translated into ALN invariants that spell out the non-increasing nature of `K_adj` and `E_adj` with respect to `rcalib` .

The second phase of the research program involves operationalization and validation. This includes implementing the core computational kernels in a production-grade language like C++ or Rust, treating them as the single source of truth for all risk calculations . A critical part of this phase is building the CI replay jobs that will serve as automated tests of the system's properties . These jobs will synthetically inject faults and verify that `rcalib`, `Vt_full`, and `R` weakly increase, while `K_adj` and `E_adj` weakly decrease. Any regression in this behavior must be sufficient to block merges, creating a robust feedback loop between empirical results and mathematical correctness. The KerWindow and KerDeployableConfig implementations must be updated to correctly incorporate `rcalib` and `rsigma` into the `R` and `E` calculations, and the associated ALN invariants must be revised to reflect these changes .

Finally, the third phase focuses on extending the framework's utility and exploring its deeper implications. This involves building user-facing tools like the proposed Ingest Quality Dashboard and Data-Trust Gatekeeper service to provide operators with actionable intelligence . It also includes exploring advanced topics, such as the co-tuning of Lyapunov weights `w_calib` and `w_sigma` to balance the influence of data quality and physical risk in the Lyapunov residual , and investigating species-specific coupling, where `rsigma` for environmental metrics could directly raise species-specific risk residuals like `V_bee` or `V_worm` . The overarching goal is to move beyond a static set of rules to a dynamic, self-validating system where data quality is an explicit, quantifiable, and enforceable component of ecological accountability. By following this path, the

framework can evolve from a promising concept into a provably safe, transparent, and indispensable tool for managing the integrity of the EcoNet ecosystem.

---

## Reference


1. The 2025 Conference on Empirical Methods in Natural Language ... https://aclanthology.org/events/emnlp-2025/
2. 2024 Index IEEE Transactions on Neural Networks and Learning ... https://ieeexplore.ieee.org/iel8/5962385/10772360/10791322.pdf
3. 动力系统/优化与控制2026_4_8 - arXiv每日学术速递 https://www.arxivdaily.com/thread/78546
4. Arxiv今日论文| 2026-01-14 - 闲记算法 http://lonepatient.top/2026/01/14/arxiv_papers_2026-01-14
5. Computer Science - arXiv https://www.arxiv.org/list/cs/new?skip=875&show=1000
6. [PDF] SCIS 控制科学精选文章目录 http://scis.scichina.com/collection/scis-control.pdf
7. [XLS] SCI https://cs.zjnu.edu.cn/_upload/article/files/95/fc/1a909bf145e0a2cf95fcf5c8136b/63e07c2a-46aa-4e04-8a4d-34bc530161ec.doc
8. Sensitivity Drift - an overview | ScienceDirect Topics https://www.sciencedirect.com/topics/engineering/sensitivity-drift
9. A Systematic Review of Advanced Sensor Technologies for Non ... https://pmc.ncbi.nlm.nih.gov/articles/PMC9965987/
10. Advanced electronic noses for future robotic olfaction - Nature https://www.nature.com/articles/s44182-025-00071-y
11. Sensors, Volume 26, Issue 2 (January-2 2026) – 402 articles - MDPI https://www.mdpi.com/1424-8220/26/2
12. Heading Drift Reduction for Foot-Mounted Inertial Navigation ... https://www.researchgate.net/publication/327198852_Heading_Drift_Reduction_for_Foot-Mounted_Inertial_Navigation_System_via_Multi-Sensor_Fusion_and_Dual-Gait_Analysis
13. Issue 2 - Volume 36 - Measurement Science and Technology https://iopscience.iop.org/issue/0957-0233/36/2
14. [PDF] Lyapunov-based uncertainty-aware safe reinforcement learning - arXiv https://arxiv.org/pdf/2107.13944

15. Learning Discrete-Time Uncertain Nonlinear Systems ... - IEEE Xplore https://ieeexplore.ieee.org/ielaam/9552933/9683993/9926168-aam.pdf
16. [PDF] Lyapunov-Stable Adaptive Control for Multimodal ... - OpenReview https://openreview.net/pdf?id=w3dMbY6rFe
17. Optimal Lp filtering for discrete − time non − Gaussian dynamic systems https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/cth2.12181
18. (PDF) Disturbance and Uncertainty Rejection Based on Fixed-Time ... https://www.researchgate.net/publication/354720779_Disturbance_and_Uncertainty_Rejection_Based_on_Fixed-Time_Sliding-Mode_Control_for_the_Secure_Communication_of_Chaotic_Systems
19. [PDF] UC San Diego - eScholarship.org https://escholarship.org/content/qt9s07n7vn/qt9s07n7vn.pdf
20. DQN-empowered energy optimization for wireless powered ... - Nature https://www.nature.com/articles/s41598-026-38904-x
21. Learnability Window in Gated Recurrent Neural Networks - arXiv https://arxiv.org/html/2512.05790v7
22. Almost-Linear RNNs Yield Highly Interpretable Symbolic Codes in ... https://arxiv.org/html/2410.14240v1
23. [PDF] Dissipative Learning: A Framework for Viable Adaptive Systems - arXiv https://arxiv.org/pdf/2601.17933
24. Hyperparameter Transfer Laws for Non-Recurrent Multi-Path Neural ... https://arxiv.org/html/2602.07494v1
25. [PDF] FAME: Adaptive Functional Attention with Expert Routing for ... - arXiv https://arxiv.org/pdf/2510.00621?
26. LoRA-DA: Data-Aware Initialization for Low-Rank Adaptation ... - arXiv https://arxiv.org/html/2510.24561v2
27. [PDF] A tensor network formalism for neuro-symbolic AI - arXiv https://arxiv.org/pdf/2601.15442
28. Circuit Complexity of Hierarchical Knowledge Tracing and ... - arXiv https://arxiv.org/html/2603.23823v1
29. [PDF] Foundations of Artificial Intelligence Frameworks: Notion and Limits ... https://arxiv.org/pdf/2511.18517
30. Redundancy as a Structural Information Principle for Learning and ... https://arxiv.org/html/2510.10938v1
31. [PDF] Lyapunov-Stable Adaptive Control for Multimodal Concept Drift https://openreview.net/pdf/1d77ccebe82a8e321702975e1d79032e6baf33d0.pdf

32. [PDF] Discrete Double-Bracket Flows for Isotropic-Noise ... - arXiv.org https://www.arxiv.org/pdf/2602.13759
33. [PDF] Interval observers-based Fault-Tolerant Control of LPV Switched ... https://theses.hal.science/tel-05184370v1/file/2025UPAST039.pdf
34. 系统与控制2026_4_2 - arXiv每日学术速递 https://arxivdaily.com/thread/78351
35. Adaptive optimal output regulation of linear discrete-time systems ... https://www.researchgate.net/publication/357382830_Adaptive_optimal_output_regulation_of_linear_discrete-time_systems_based_on_event-triggered_output-feedback
36. Absolute stability http://ieeexplore.ieee.org/iel5/9/21042/00975523.pdf
37. [PDF] An Invariant Compiler for Neural ODEs in AI-Accelerated Scientific ... https://arxiv.org/pdf/2603.23861
38. [PDF] Course Introduction for International Postgraduates School of ... https://global.buaa.edu.cn/en/Course_Introduction_for_International_Postgraduates.pdf
39. CASE: Deterministic Supervisory Stability Governor for Complex ... https://www.linkedin.com/posts/andre-watts-a0a076321_case-formal-stability-governor-specification-activity-7431492777475158016-wl1n
40. Sensors, Volume 25, Issue 24 (December-2 2025) – 289 articles https://www.mdpi.com/1424-8220/25/24
41. Autonomous Agents on Blockchains: Standards, Execution Models ... https://arxiv.org/html/2601.04583v1
42. [PDF] The 10th Workshop on Financial Technology and Natural Language ... https://aclanthology.org/2025.finnlp-2.pdf
43. [PDF] Proof-of-State for Financial Solvency, Liquidity, and ... - SEC.gov https://www.sec.gov/files/ctf-written-input-proof-state-financial-solvency-liquidity-economic-exposure-costa-031226.pdf
44. Sensors, Volume 26, Issue 3 (February-1 2026) – 328 articles - MDPI https://www.mdpi.com/1424-8220/26/3
45. Sensors, Volume 26, Issue 7 (April-1 2026) – 273 articles https://www.mdpi.com/1424-8220/26/7
46. Sensors, Volume 26, Issue 6 (March-2 2026) – 271 articles https://www.mdpi.com/1424-8220/26/6
47. Sensors, Volume 25, Issue 14 (July-2 2025) – 286 articles https://www.mdpi.com/1424-8220/25/14
48. Sensors, Volume 26, Issue 5 (March-1 2026) – 338 articles https://www.mdpi.com/1424-8220/26/5

49. Sensors, Volume 25, Issue 9 (May-1 2025) – 322 articles https://www.mdpi.com/1424-8220/25/9
50. Sensors, Volume 25, Issue 17 (September-1 2025) – 392 articles https://www.mdpi.com/1424-8220/25/17
51. Sensors, Volume 25, Issue 13 (July-1 2025) – 405 articles https://www.mdpi.com/1424-8220/25/13
52. [PDF] Governance Architecture for Autonomous Agent Systems - arXiv https://arxiv.org/pdf/2603.07191
53. [PDF] 2025 Annual Report - Huawei https://www-file.huawei.com/admin/asset/v1/pro/view/dec782afd1544f558c32fc76e3141334.pdf
54. [XLS] China https://www.nsfc.gov.cn/Portals/0/fj/fj20220316_02.xlsx
55. Response-Aware Risk-Constrained Control Barrier Function ... - arXiv https://arxiv.org/html/2603.24598v1
56. Steering Robots with Inference-Time Interactions - arXiv https://arxiv.org/html/2506.14287v1
57. [PDF] A Historical Perspective of Adaptive Control and Learning - arXiv https://arxiv.org/pdf/2108.11336
58. [PDF] Why Alignment Needs Formal Control Theory (and an ... - arXiv https://arxiv.org/pdf/2506.17846
59. [PDF] Solving Neural Min-Max Games: The Role of Architecture ... - arXiv https://arxiv.org/pdf/2512.00389
60. [PDF] arXiv:2502.04662v1 [cs.LG] 7 Feb 2025 https://arxiv.org/pdf/2502.04662
61. Adversarial Vulnerabilities in Neural Operator Digital Twins - arXiv https://arxiv.org/html/2603.22525v1
62. A Functional-Analytic Framework for Nonlinear Adaptive Memory https://arxiv.org/html/2604.03852v1
63. [PDF] Adaptive Coverage Policies in Conformal Prediction - arXiv https://arxiv.org/pdf/2510.04318
64. Automated Circuit Discovery with Provable Guarantees - arXiv https://arxiv.org/html/2602.16823v1
65. [PDF] Robust Training of Neural Networks using Scale Invariant ... - arXiv https://arxiv.org/pdf/2202.00980
66. [PDF] Multivariate Conformal Selection - arXiv https://arxiv.org/pdf/2505.00917
67. [PDF] A Variational Framework for Residual-Based Adaptivity in Neural ... https://arxiv.org/pdf/2509.14198?

68. [PDF] A provable control of sensitivity of neural networks through a direct ... https://arxiv.org/pdf/2404.09821
69. [PDF] Provably Minimum-Length Conformal Prediction Sets for Ordinal ... https://arxiv.org/pdf/2511.16845
70. [PDF] Forward Invariance in Neural Network Controlled Systems - arXiv https://arxiv.org/pdf/2309.09043
71. [PDF] DIGRAF: Diffeomorphic Graph-Adaptive Activation Function - arXiv https://arxiv.org/pdf/2407.02013
72. Categorical framework for quantum-resistant zero-trust AI security https://www.nature.com/articles/s41598-026-37190-x
73. Mathematics, Volume 14, Issue 3 (February-1 2026) – 194 articles https://www.mdpi.com/2227-7390/14/3
74. Calibration of metallicity of LAMOST M dwarf stars using FGK+ ... https://academic.oup.com/mnras/article/527/4/11866/7503365
75. MAGELLAN ADAPTIVE OPTICS FIRST-LIGHT ... https://iopscience.iop.org/article/10.1088/0004-637X/815/2/108
76. SCIS Collection of LETTER - SCIENCE CHINA Information Sciences http://scis.scichina.com/scis-letter.html
77. Sensors, Volume 25, Issue 21 (November-1 2025) – 306 articles https://www.mdpi.com/1424-8220/25/21
78. Sensors, Volume 25, Issue 5 (March-1 2025) – 359 articles - MDPI https://www.mdpi.com/1424-8220/25/5
79. Sensors, Volume 23, Issue 16 (August-2 2023) – 295 articles https://www.mdpi.com/1424-8220/23/16
80. Adaptive Fault-Tolerant Tracking Control for Discrete-Time ... https://www.researchgate.net/publication/341149289_Adaptive_Fault-Tolerant_Tracking_Control_for_Discrete-Time_Multiagent_Systems_via_Reinforcement_Learning_Algorithm
81. Mathematics, Volume 13, Issue 24 (December-2 2025) – 149 articles https://www.mdpi.com/2227-7390/13/24
82. Electrical Engineering and Systems Science - arXiv https://arxiv.org/list/eess/new
83. Issue 9 - Volume 100 - Physica Scripta - IOPscience https://iopscience.iop.org/issue/1402-4896/100/9
84. Sensors, Volume 25, Issue 12 (June-2 2025) – 264 articles https://www.mdpi.com/1424-8220/25/12
85. [PDF] UC San Diego Electronic Theses and Dissertations - eScholarship.org https://escholarship.org/content/qt1630b2mk/qt1630b2mk.pdf

86. [DOC] 20110000 - | European Securities and Markets Authority https://www.esma.europa.eu/sites/default/files/esma_mifid2_dp_fese_replyform_1.doc
87. [PDF] EMISSIONS TRADING IN PRACTICE: https://icapcarbonaction.com/system/files/document/icap-pmr_ets_handbook_1st_edition.pdf
88. [PDF] OECD Economic Surveys: Indonesia 2016 (EN) https://www.oecd.org/content/dam/oecd/en/publications/reports/2016/10/oecd-economic-surveys-indonesia-2016_g1g71059/eco_surveys-idn-2016-en.pdf
89. [PDF] Lessons from NAFTA for Latin America and the Caribbean https://publications.iadb.org/publications/english/document/Lessons-from-NAFTA-For-Latin-America-and-the-Caribbean.pdf
90. multi0page.txt - Documents & Reports - World Bank https://documents1.worldbank.org/curated/en/459951468280169927/txt/multi0page.txt
91. Sensors, Volume 24, Issue 13 (July-1 2024) – 395 articles https://www.mdpi.com/1424-8220/24/13
92. 无老师Toefl 词汇红宝书- 7 天搞定托福单词(PDFDrive) - Scribd https://www.scribd.com/document/799989449/%E6%97%A0%E8%80%81%E5%B8%88Toefl-%E8%AF%8D%E6%B1%87%E7%BA%A2%E5%AE%9D%E4%B9%A6-7-%E5%A4%A9%E6%90%9E%E5%AE%9A%E6%89%98%E7%A6%8F%E5%8D%95%E8%AF%8D-PDFDrive
93. [PDF] Open Source Used In CNAAP Standard Service Assurance 1.0 - Cisco https://www.cisco.com/c/dam/en_us/about/doing_business/open_source/docs/CNAAPStandardServiceAssurance-10-1678791766.pdf
94. (PDF) Sanjeevani Advanced Learner's Dictionary - Academia.edu https://www.academia.edu/43276191/Sanjeevani_Advanced_Learners_Dictionary
95. Arxiv今日论文| 2026-02-05 - 闲记算法 http://lonepatient.top/2026/02/05/arxiv_papers_2026-02-05
96. 机器学习2026_1_13 - arXiv每日学术速递 http://arxivdaily.com/thread/75632
97. 289110V2.txt - Documents & Reports - World Bank https://documents1.worldbank.org/curated/en/276151468235735884/txt/289110V2.txt
98. Literacy and the possible dream - UNESCO Digital Library https://unesdoc.unesco.org/ark:/48223/pf0000018354
99. Arxiv今日论文| 2026-03-11 - 闲记算法 http://lonepatient.top/2026/03/11/arxiv_papers_2026-03-11
100. (PDF) Digital Twins in Intelligent Transportation and Communication ... https://www.researchgate.net/publication/403404606_Digital_Twins_in_Intelligent_Transportation_and_Communication_Systems_A_Survey

101. [PDF] Learning Stable Koopman Embeddings for Identification and Control https://arxiv.org/pdf/2401.08153
102. [PDF] Curricula Syllabus for International Postgraduates (Taught in ... https://global.buaa.edu.cn/en/Beihang_Course_Description_for_ALL_Master_Programs_2021.pdf
103. [PDF] Sampling Techniques for Computational Statistical Physics https://ins.sjtu.edu.cn/people/shijin/PS/EACM.pdf
104. [PDF] Chaotic dynamics and partial hyperbolicity https://theses.hal.science/tel-01948392v1/document
105. [PDF] TBSI course introduction-detail Update June 2020 https://www.sigs.tsinghua.edu.cn/_upload/article/files/22/55/befbd7c94a719dc3c46abe1e9c9b/73bc2fac-a7ea-4e38-a2f8-608007bd3b18.pdf
106. Sensors, Volume 25, Issue 8 (April-2 2025) – 292 articles https://www.mdpi.com/1424-8220/25/8
107. Sensors, Volume 25, Issue 15 (August-1 2025) – 353 articles https://www.mdpi.com/1424-8220/25/15
108. Sensors, Volume 25, Issue 18 (September-2 2025) – 326 articles https://www.mdpi.com/1424-8220/25/18
109. The Glossary of Prosthodontics Terms 9 Edition GPT9 | PDF - Scribd https://www.scribd.com/document/357268711/The-Glossary-of-Prosthodontics-Terms-9-Edition-GPT9
110. Cambridge Advanced Learners Dictionary | PDF | Verb | Adjective https://www.scribd.com/document/748647350/Colin-McIntosh-Ed-Cambridge-Advanced-Learner-s-Dictionary-Cambridge-University-Press-2013
111. Findings of the Association for Computational Linguistics: EACL 2026 https://aclanthology.org/volumes/2026.findings-eacl/
112. [PDF] Food Security in Africa - FAO Knowledge Repository https://openknowledge.fao.org/server/api/core/bitstreams/4b401528-00e8-4539-bcb0-d99ffb3f901b/content
113. A Review On Safe Reinforcement Learning Using Lyapunov ... - arXiv https://arxiv.org/html/2508.09128v1
114. [PDF] Zubov-Net: Adaptive Stability for Neural ODEs Reconciling Accuracy ... https://arxiv.org/pdf/2509.21879
115. Neural Lyapunov Control for Discrete-Time Systems - arXiv https://arxiv.org/html/2305.06547v3
116. Safe and Stable Control via Lyapunov-Guided Diffusion Models - arXiv https://arxiv.org/html/2509.25375v1

117. Neural Vector Lyapunov–Razumikhin Certificates for Delayed ... https://arxiv.org/html/2604.00774v1

118. Constructive Lyapunov Functions via Topology-Preserving Neural ... https://arxiv.org/html/2510.24730v1

119. Two-Stage Learning of Stabilizing Neural Controllers via Zubov ... https://arxiv.org/html/2506.01356v1

120. [PDF] Neural Lyapunov Control for Discrete-Time Systems - arXiv https://arxiv.org/pdf/2305.06547

121. Hebbian-Oscillatory Co-Learning - arXiv https://arxiv.org/html/2603.08731

122. IEEE_Sensors22__NUS_FM_A... https://laas.hal.science/hal-03925513v1/preview/IEEE_Sensors22__NUS_FM_AM_OFI_HAL.pdf

123. Sensors, Volume 25, Issue 10 (May-2 2025) – 295 articles https://www.mdpi.com/1424-8220/25/10

124. From Logic Monopoly to Social Contract: Separation of Power and ... https://arxiv.org/html/2603.25100v1

125. [PDF] united nations conference on trade and development https://digitallibrary.un.org/record/559213/files/ditcclp20052_en.pdf

126. multi-page.txt - World Bank Documents https://documents1.worldbank.org/curated/en/264801468760536273/txt/multi-page.txt

127. [PDF] Current status and future development of modular high temperature ... https://www-pub.iaea.org/MTCD/Publications/PDF/te_1198_prn.pdf

128. [PDF] Stochastic Approximation Beyond Gradient for Signal Processing ... https://arxiv.org/pdf/2302.11147

129. [PDF] Machine Learning with Physics Knowledge for Prediction: A Survey https://arxiv.org/pdf/2408.09840

130. [PDF] Optimal guidance and estimation of a 2D diffusion-advection ... - arXiv https://arxiv.org/pdf/2111.01343

131. [PDF] Building causation links in stochastic nonlinear systems from data https://arxiv.org/pdf/2509.07701?

132. [PDF] Noisy Recurrent Neural Networks - arXiv https://arxiv.org/pdf/2102.04877

133. [PDF] Safe and Robust Domains of Attraction for Discrete-Time Systems https://arxiv.org/pdf/2603.03082

134. Two-Stage Learning of Stabilizing Neural Controllers via Zubov ... https://arxiv.org/html/2506.01356v2

135. [PDF] &ISSILE 3TRATEGIES 3USTAINABLE https://www-pub.iaea.org/MTCD/Publications/PDF/P1288_web.pdf

136. [PDF] Implementing Domestic Tradable Permits for Environmental Protection https://www.oecd.org/content/dam/oecd/en/publications/reports/1999/08/implementing-domestic-tradable-permits-for-environmental-protection_g1gh2818/9789264181182-en.pdf
137. 69256-REVISED-ENGLISH-PUBLIC-RR-Toolkit-EN-New-report ... https://documents1.worldbank.org/curated/en/529921469672181559/txt/69256-REVISED-ENGLISH-PUBLIC-RR-Toolkit-EN-New-report-date-2017-12-27.txt
138. [PDF] Neural Vector Lyapunov-Razumikhin Certificates for Delayed ... - arXiv https://arxiv.org/pdf/2604.00774
139. [PDF] arXiv:2210.16940v4 [cs.LG] 22 Dec 2023 https://arxiv.org/pdf/2210.16940
140. General Machine Learning Theory for Learning Under Variable ... https://arxiv.org/html/2603.23220v1
141. Verification of Neural Control Barrier Functions with Symbolic ... - arXiv https://arxiv.org/html/2410.16281v1
142. Arxiv今日论文| 2026-03-20 - 闲记算法 http://lonepatient.top/2026/03/20/arxiv_papers_2026-03-20
143. Sensors, Volume 23, Issue 2 (January-2 2023) – 492 articles https://www.mdpi.com/1424-8220/23/2
144. (PDF) Koopman-Based Control System for Quadrotors in Noisy ... https://www.researchgate.net/publication/380749319_Koopman-based_Control_System_for_Quadrotors_in_Noisy_Environments
145. HSK Exam Preparation Guide | PDF | Linguistics | Grammar - Scribd https://www.scribd.com/document/869622956/Hsk-Standard-Course-3-Textbook-compressed-1
146. Arxiv今日论文| 2026-03-11 - 闲记算法 http://lonepatient.top/2026/03/11/arxiv_papers_2026-03-11.html
147. Quantum-enhanced privacy aggregation for healthcare monitoring ... https://www.nature.com/articles/s41598-026-43649-8_reference.pdf
148. [PDF] Security Considerations for Multi-agent Systems - arXiv https://arxiv.org/pdf/2603.09002
149. Data and statistics: empirical bases for the social sciences https://unesdoc.unesco.org/ark:/48223/pf0000116424
150. Utilizing reinforcement learning in feedback control of nonlinear ... https://www.sciencedirect.com/science/article/pii/S2772508125000614
151. On the use of a Euclidean norm function for the estimation ... - PubMed https://pubmed.ncbi.nlm.nih.gov/27461568/

152. [PDF] Sensor-based control of a heterogeneous and autonomous multi ... https://theses.hal.science/tel-05539669v2/file/2025COAZ4034.pdf

153. Mathematics, Volume 10, Issue 19 (October-1 2022) – 308 articles https://www.mdpi.com/2227-7390/10/19

154. (PDF) MME-based attitude dynamics identification and estimation for ... https://www.academia.edu/31833435/MME_based_attitude_dynamics_identification_and_estimation_for_SAMPEX

155. Arxiv今日论文| 2026-02-19 - 闲记算法 http://lonepatient.top/2026/02/19/arxiv_papers_2026-02-19

156. [PDF] Risk management and decision-making in relation to sustainable ... https://www.ipcc.ch/site/assets/uploads/sites/4/2019/11/10_Chapter-7.pdf

157. [PDF] I PHOTOCOPYING DERIVATIVE WORKS ELECTRONIC STORAGE ... https://www.researchgate.net/profile/Emad-Rabboh/publication/331654855_The_Correlation_of_Deconstruction_Architecture_to_Arab_Architectural_Identity/links/5cfcbf1b92851c874c59a0e0/The-Correlation-of-Deconstruction-Architecture-to-Arab-Architectural-Identity.pdf

158. Morning Briefing - 2025 • Yardeni Research https://archive.yardeni.com/morning-briefing-2025/

159. Sensors, Volume 19, Issue 18 (September-2 2019) – 242 articles https://www.mdpi.com/1424-8220/19/18

160. Latest articles - MDPI https://www.mdpi.com/latest_articles

161. Appl. Sci., Volume 14, Issue 18 (September-2 2024) – 480 articles https://www.mdpi.com/2076-3417/14/18

162. Computational issues in biaxial bending capacity assessment of RC ... https://www.sciencedirect.com/science/article/pii/S0045794924002062

163. Measuring Complexity at the Requirements Stage: Spectral Metrics ... https://www.mdpi.com/2079-8954/14/4/364

164. [PDF] LMI Optimization Based Multirate Steady-State Kalman Filter Design https://arxiv.org/pdf/2602.01537

165. A Novel Decoupled Synchronous Control Method for Multiple ... - MDPI https://www.mdpi.com/2076-3417/12/15/7551

166. [PDF] Introduction to Optimal Control - HAL https://cel.hal.science/hal-02987731v2/document

167. Sampled-data-based fault-tolerant consensus control for multi-agent ... https://www.sciencedirect.com/science/article/abs/pii/S0005109821003678

168. 机器学习2026_1_1 - arXiv每日学术速递 http://arxivdaily.com/thread/75279

169. [PDF] Efficient Distributed Optimization under Heavy-Tailed Noise - arXiv https://arxiv.org/pdf/2502.04164

170. [PDF] UCLA Electronic Theses and Dissertations - eScholarship.org https://escholarship.org/content/qt0hq9b2rc/qt0hq9b2rc.pdf

171. [PDF] Mathematical Foundations of Deep Learning - HAL https://hal.science/hal-04928560v2/file/Sourangshu_Ghosh_IISc_Bangalore_Mathematical_Foundations_of_Deep_Learning_Version_2.pdf

172. Distributed Identification of Contracting and/or Monotone Network ... https://ieeexplore.ieee.org/iel7/9/4601496/09547745.pdf

173. [PDF] optimization under uncertainty | castle https://castle.princeton.edu/wp-content/uploads/2017/09/Powell_OptimizationUnderUncertaintySept142017.pdf

174. [PDF] Stochastic Approximation Beyond Gradient for Signal ... - HAL https://hal.science/hal-04174351/document

175. from Hybrid Zero Dynamics to Control Lyapunov Functions Via ... https://search.proquest.com/openview/491b0f122fb7c36f989589731847958d/1?pq-origsite=gscholar&cbl=18750&diss=y

176. Arxiv今日论文| 2026-04-03 - 闲记算法 http://lonepatient.top/2026/04/03/arxiv_papers_2026-04-03

177. 机器学习2026_4_2 - arXiv每日学术速递 https://arxivdaily.com/thread/78368

178. Arxiv今日论文| 2026-03-04 - 闲记算法 http://lonepatient.top/2026/03/04/arxiv_papers_2026-03-04.html

179. Fugu-MT: arxivの論文翻訳(概要) https://fugumt.com/fugumt/paper/index.html

180. [PDF] Quasi-optimal Reinforcement Learning with Continuous Actions - arXiv https://arxiv.org/pdf/2301.08940

181. [PDF] Mathematical Foundations of Deep Learning | HAL https://hal.science/hal-04928560v5/file/Sourangshu_Ghosh_IISc_Bangalore_Mathematical_Foundations_of_Deep_Learning_Version_7%20%288%29.pdf

182. Online Optimization of Dynamical Systems With Deep Learning ... https://ieeexplore.ieee.org/iel7/9552933/9683993/09891838.pdf

183. Electronics, Volume 14, Issue 22 (November-2 2025) – 195 articles https://www.mdpi.com/2079-9292/14/22

184. Robust Reasoning as a Symmetry-Protected Topological Phase https://arxiv.org/html/2601.05240v1

185. A Recursive Approach for Magnetic Field Estimation in Spacecraft ... https://www.mdpi.com/2226-4310/9/12/833

186. A Review of Modeling and Control Techniques for Unmanned Aerial ... https://onlinelibrary.wiley.com/doi/full/10.1002/eng2.70215

187. [PDF] Contact-Aided State Estimation on Lie Groups for Legged Robot ... https://grizzle.robotics.umich.edu/files/dissertation_ross.pdf

188. [PDF] An Advanced Exploration of Reinforcement Learning for Mobile ... https://arxiv.org/pdf/2404.14238

189. [PDF] A Generative Framework for MIMO Channel Estimation - arXiv https://arxiv.org/pdf/2601.15767

190. Physics-informed machine learning: A comprehensive review on ... https://www.sciencedirect.com/science/article/pii/S0957417424015458

191. Optimized topology control for large-scale IoT networks using graph ... https://www.nature.com/articles/s41598-026-43621-6_reference.pdf

192. The centrality of population-level factors to network computation is ... https://pmc.ncbi.nlm.nih.gov/articles/PMC10118067/

193. Exploring the Potential of Topological Data Analysis for Explainable ... https://www.mdpi.com/2227-7390/14/2/378

194. Full article: Geospatial Sentiment Analytics for Hospitality Management https://www.tandfonline.com/doi/full/10.1080/1528008X.2026.2639380

195. Hybrid Fusion Paradigm in Advanced Process Monitoring https://pubs.acs.org/doi/10.1021/acs.iecr.5c02759

196. [PDF] Metric Convolutions: A Unifying Theory to Adaptive Image ... https://openaccess.thecvf.com/content/ICCV2025/papers/Dages_Metric_Convolutions_A_Unifying_Theory_to_Adaptive_Image_Convolutions_ICCV_2025_paper.pdf

197. [PDF] On the Stability of Neural Networks in Deep Learning - HAL theses https://theses.hal.science/tel-05398597v1/file/2025UPSLD022.pdf

198. Sensors, Volume 25, Issue 19 (October-1 2025) – 327 articles https://www.mdpi.com/1424-8220/25/19

199. [PDF] explicit construction of quadratic lyapunov functions for the small ... http://dsbaero.engin.umich.edu/wp-content/uploads/sites/441/2020/06/ExplicitQuadraticPart2DT.pdf

200. Distributed Consensus Control for Discrete-Time T–S Fuzzy Multiple ... https://pmc.ncbi.nlm.nih.gov/articles/PMC11679411/

201. Resilient Event-Triggered Observer-Based Control of Uncertain ... https://ieeexplore.ieee.org/iel8/6287639/10820123/11149657.pdf

202. [PDF] Robust H∞ Observer Design via Finsler's Lemma and IQCs - arXiv https://arxiv.org/pdf/2604.03989

203. Mathematics, Volume 14, Issue 5 (March-1 2026) – 181 articles - MDPI https://www.mdpi.com/2227-7390/14/5

204. [PDF] Risk-averse autonomous systems - arXiv https://arxiv.org/pdf/2109.08947

205. [PDF] Mathematical Foundations of Trustworthy AI - HAL https://hal.science/hal-05449393v1/document

206. A Review of Modeling and Control Techniques for Unmanned Aerial ... https://onlinelibrary.wiley.com/doi/10.1002/eng2.70215

207. Multi-Sensor Adaptive Weighted Data Fusion Based on Biased ... https://pmc.ncbi.nlm.nih.gov/articles/PMC11174378/

208. (PDF) Multi-Sensor Adaptive Weighted Data Fusion Based on ... https://www.researchgate.net/publication/380765355_Multi-Sensor_Adaptive_Weighted_Data_Fusion_Based_on_Biased_Estimation

209. [PDF] Sensor fault detection for switched systems using interval observer ... https://www.sciencedirect.com/science/article/am/pii/S0947358020301126

210. [PDF] Zonotopic observer designs for uncertain Takagi–Sugeno fuzzy ... https://uphf.hal.science/hal-04278847/document

211. [PDF] cooperative observer-based h∞ fault-tolerant tracking control ... - arXiv https://arxiv.org/pdf/2604.03921

212. [PDF] Towards a Turnkey Model Predictive Controller - eScholarship.org https://escholarship.org/content/qt9q0155dd/qt9q0155dd.pdf

213. Appl. Sci., Volume 16, Issue 7 (April-1 2026) – 502 articles https://www.mdpi.com/2076-3417/16/7

214. Neural Information Processing - Springer https://link.springer.com/content/pdf/10.1007/978-981-99-8079-6.pdf